



Next Generation of Policy Frameworks and Modeling

WEEK 2 · JUNE 15-19, 2026

Risk-Adjusted UIP

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Domestic Interest Rate Compensation for Expected Depreciation (in log terms) Foreign Interest Rate Country Risk Premium

↓ ↓ ↓ ↓

$i_t = [s_{t+1} - s_t] + [i_t^f + u_t]$

Derive Multi-Period Risk Adjusted UIP

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$$i_{t+1} = [s_{t+2} - s_{t+1}] + [i_{t+1}^f + u_{t+1}]$$

Derive Multi-Period Risk Adjusted UIP

$$\begin{aligned}i_t &= [s_{t+1} - s_t] + [i_t^f + u_t] \\i_{t+1} &= [s_{t+2} - s_{t+1}] + [i_{t+1}^f + u_{t+1}] \\&\vdots \\i_{t+k} &= [s_{t+k+1} - s_{t+k}] + [i_{t+k}^f + u_{t+k}]\end{aligned}$$

$$\sum_{j=0}^k i_{t+j} = [s_{t+k+1} - s_t] + \sum_{j=0}^k [i_{t+j}^f + u_{t+j}]$$

Derive Multi-Period Risk Adjusted UIP

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$$\sum_{j=0}^k i_{t+j} = [s_{t+k+1} - s_t] + \sum_{j=0}^k [i_{t+j}^f + u_{t+j}]$$

Important Insights from Risk-Adjusted UIP

Expected policy
rate path

Expected
depreciation

Expected foreign policy rate path
adjusted for country risk premium

$$\sum_{j=0}^k i_{t+j} = [s_{t+k+1} - s_t] + \sum_{j=0}^k [i_{t+j}^f + u_{t+j}]$$

Expected real
policy rate path

Expected real
depreciation

Expected real foreign policy rate path
adjusted for country risk premium

$$\sum_{j=0}^k r_{t+j} = [z_{t+k+1} - z_t] + \sum_{j=0}^k [r_{t+j}^f + u_{t+j}]$$

Important Insights from Risk-Adjusted UIP

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$$\sum_{j=0}^k r_{t+j} = [z_{t+k+1} - z_t] + \sum_{j=0}^k [r_{t+j}^f + u_{t+j}]$$

Market Interest Rates and Exchange Rates As Shock Absorbers

Negative demand shock reduces inflation in the short run. IFT CB expected to reduce the policy rate sufficiently to steer inflation back to target.

$$\downarrow \sum_{j=0}^k i_{t+j} = [s_{t+k+1} - \uparrow s_t] + \sum_{j=0}^k [i_{t+j}^f + u_{t+j}]$$

Nominal exchange rate also depreciates today. Easier monetary conditions coming from both interest rates and exchange rates.

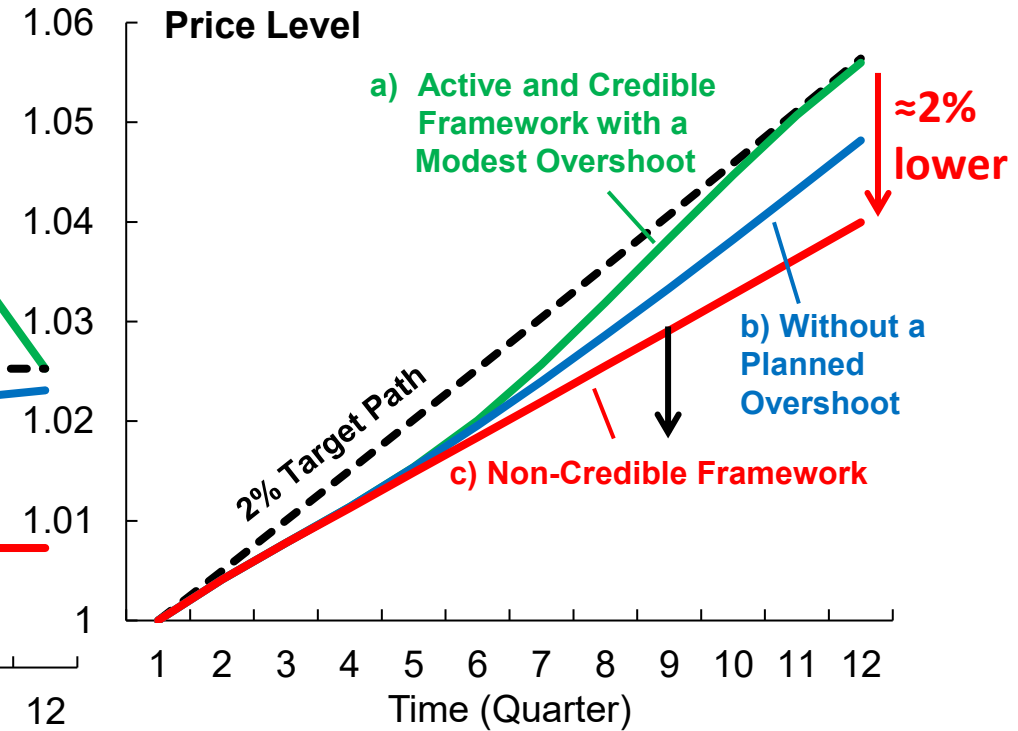
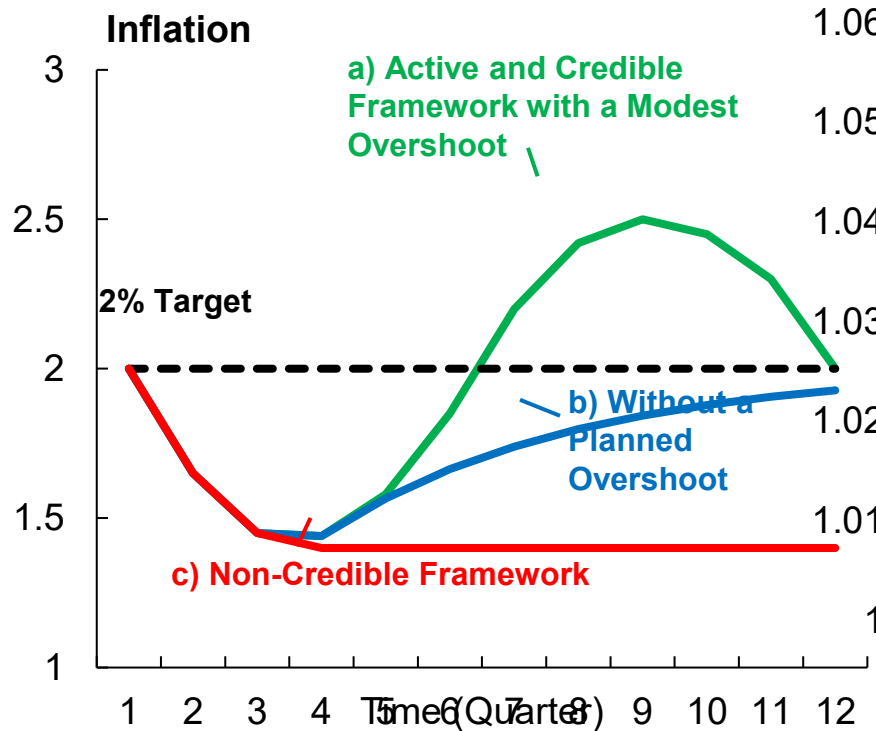
An important assumption: nominal exchange rate in the long-run does not change. Under what conditions will it happen?

[Expected Domestic Interest Rates = Expected Nominal Depreciation + Expected Foreign Interest Rates Plus Risk Premium]

$$\sum_{j=0}^k i_{t+j} = \underbrace{[s_{t+k+1} - s_t]}_{\substack{\downarrow \\ \text{a) } -200 \text{ bps} \\ \text{b) } -100 \text{ bps} \\ \text{c) } 0 \text{ bps}}} + \sum_{j=0}^k [i_{t+j}^f + u_{t+j}]$$

- | | | | | | |
|----|----------|---|--------|---|---|
| a) | -200 bps | = | 0 % | - | (2 %) (significant depreciation today) |
| b) | -100 bps | = | -0.8 % | - | (0.2 %) (less depreciation today) |
| c) | 0 bps | = | -2 % | - | (-2 %) (significant appreciation today) |

Notice the sign flips!



Market Interest Rates and Exchange Rates As Shock Amplifiers

Negative demand shock reduces inflation in the short run. When the nominal interest rate is constrained by the ZLB, inflation can ratchet downwards, causing nominal exchange rate to appreciate.

$$\sum_{j=0}^k i_{t+j} = \downarrow [s_{t+k+1} - \downarrow s_t] + \sum_{j=0}^k [i_{t+j}^f + u_{t+j}]$$

Real interest rate also rises which contracts the economy even further.

$$\uparrow r_t = i_t + \downarrow E_t \pi_{t+1}$$

Real Exchange Rate

$$Z_t = S_t + P_t^f - P_t^d$$

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$$Z_t = S_t + P_t^f - P_t^d$$

$$Z_{ss} = S_{ss} + P_{ss}^f - P_{ss}^d$$

Real Exchange Rate

$$Z_t = S_t + P_t^f - P_t^d$$

$$Z_{TC} = S_{TC} + P_{TC}^f - P_{TC}^d \downarrow$$

Real Exchange Rate

$$Z_t = S_t + P_t^f - P_t^d$$

$$Z_{TC} = S_{TC} \downarrow + P_{TC}^f - P_{TC}^d \downarrow$$

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Example of Shock Amplifier

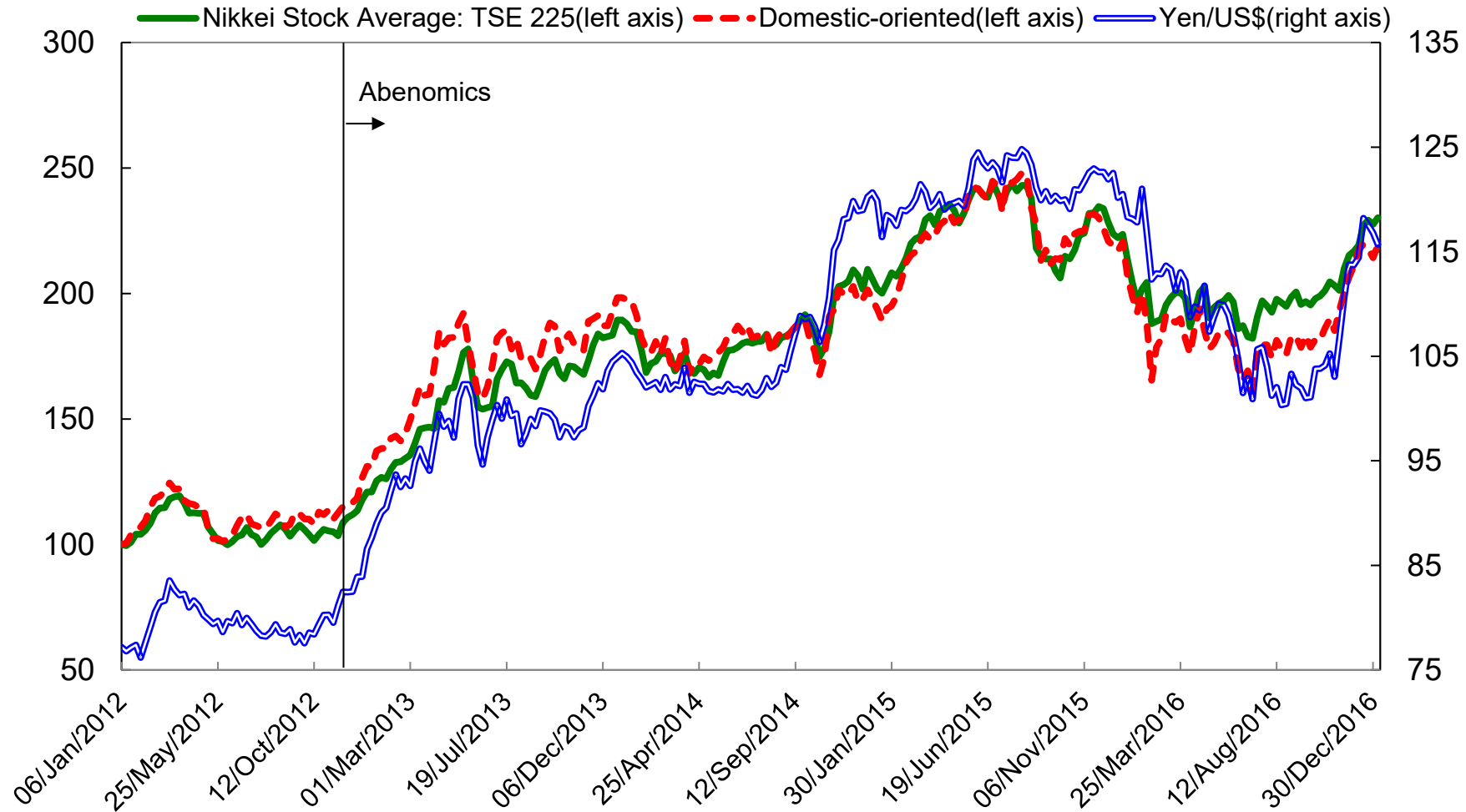
Canada versus Japan after the Global Financial Crisis

	Real GDP Growth	Real Export Growth	Output Gap	CPI Inflation	3-yr Ahead Inflation Expectation	Long-term Interest Rate	Short-term Interest Rate	Bilateral USD Exchange Rate
Canada								
2007	2.0	1.1	1.9	2.1	2.1	4.3	4.2	100.0
2008	1.2	-4.5	0.9	2.4	2.1	3.6	2.4	99.3
2009	-2.7	-13.1	-3.5	0.3	2.1	3.2	0.4	106.4
Japan								
2007	2.2	8.7	0.4	0.1	0.9	1.7	0.6	100.0
2008	-1.0	1.4	-1.4	1.4	0.9	1.5	0.4	87.8
2009	-5.5	-24.2	-7.1	-1.3	0.3	1.4	0.1	79.5

Note: Numbers in the table are in percent. Bilateral USD exchange rate is normalized to 100 in 2007.
Source: IMF World Economic Outlook October 2014.

Example of Shock Amplifier

Japanese Yen Appreciates on Bad News



Source: Haver Analytics.



Great Things are Coming.



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